

Transmission Expansion and Reliability in Peru's Energy Transition: A System Dynamics Approach

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Abstract

Peru combines abundant solar, wind and hydro resources with a generation mix still built on hydropower and natural gas, a longitudinal grid that links three load zones in series, and a planning process in which generation and transmission projects advance on very different timescales. Which of these uncertainties actually governs the system's trajectory is not obvious, and it matters for where planning effort should go. We build a system dynamics model of the Peruvian power system, coupled to a congestion-aware dispatch layer, and simulate it monthly from 2025 to 2050 over a designed experiment of 129 runs: a central case, eight one-at-a-time sensitivity runs, and 120 Latin-hypercube Monte Carlo samples of four uncertain drivers — fuel prices, demand growth, and project lead times on the generation and the transmission side. The result is unambiguous and, we argue, policy-relevant: demand growth dominates every outcome, with Spearman rank correlations of 0.99 for transmission length, 0.97 for installed capacity and -0.75 for the average price, while the other three drivers stay below 0.27 in absolute value. A $\pm 20\%$ band on the 2050 demand level moves required transmission by -18.9% to $+23.8\%$ and installed capacity by -24.0% to $+29.3\%$. The price moves the opposite way to intuition: higher demand lowers the long-run average price, because it pulls forward low-cost renewable capacity that the central case defers. Reporting a central, a best and a worst case on a single declared metric — the average delivered price — shows that the least-cost outcome is also the most transmission-intensive, requiring 26 % more network than the central case. For Peru, the binding planning question is therefore not how to protect the system against fuel-price risk but whether the grid can be built fast enough to serve demand that is itself the main source of uncertainty.

Keywords: Power system expansion planning, System dynamics, Transmission expansion, Uncertainty analysis, Latin hypercube sampling, Peru

1. Introduction

Expansion planning rests on numbers nobody knows. What fuel will cost a decade out, how fast demand will grow, and how late projects will actually come online are all uncertain, and each of them is routinely handled by picking a central value and running a scenario around it. That practice is defensible only if the uncertainties are of comparable weight. When one of them dominates, effort spent hedging the others is effort misallocated, and the planning question changes from *what might happen* to *which unknown is worth managing*.

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Peru is a useful place to ask that question, for three reasons. First, the generation mix is still built on hydropower and natural gas, with non-conventional renewables at roughly 5 % of generation despite a resource base that modelling work repeatedly finds economically attractive [14, 10]. Second, the transmission network is longitudinal: three load zones strung in series along the coast and the Andes, with the best wind and solar resources systematically distant from the demand they would serve [51]. Third, the institutional record is mixed in an instructive way. The 1992 restructuring produced genuine competition and improved security of supply [38, 34], and Peru was among the regional pioneers of technology-specific renewable auctions; but the auction mechanism has since been suspended, leaving developers without a reliable off-take signal, and new investment in both generation and transmission slowed once the initial wave of privatisation ran its course. The country therefore holds, simultaneously, a strong resource endowment, a grid that was not built for it, and a planning process whose delivery times are themselves uncertain.

Most of the modelling literature addresses this kind of problem with optimisation. Generation expansion planning, transmission expansion planning and their joint formulation are mature fields with a large and still-growing body of work [18, 4, 20, 49, 52]. These models are indispensable for identifying what an efficient expansion path would look like, but they answer a normative question — what *should* be built — under an assumption of perfect rationality and a single optimising agent. The system they describe is not the system that exists. In a liberalised market, generation investment is decentralised and driven by expected profitability, transmission is planned centrally and built on a different and longer clock, and the gap between the two is bridged by neither [12, 21]. It is precisely in that gap that delays, congestion and stranded capacity appear.

System dynamics was developed for exactly this situation: feedback, accumulation and delay [17, 47]. Applied to electricity it has a long record of reproducing the investment cycles, capacity gaps and price swings that optimisation abstracts away [16, 2, 3, 55], and recent work continues to use it to evaluate policy in liberalised markets [26, 24, 32]. What it is less often asked to do is to quantify, rather than illustrate, the relative weight of the uncertainties that drive it. A single simulation — or a handful of named scenarios — shows what the model does; it does not show which input the answer actually depends on.

This paper closes that gap for Peru. We build a system dynamics model of the Peruvian power system, coupled to a congestion-aware dispatch layer, with an explicit six-stage regulatory and construction pipeline for both generation and transmission projects. We then subject it to a designed experiment of 129 simulations: a central case, eight one-at-a-time runs that isolate each driver, and 120 Latin-hypercube Monte Carlo samples in which four drivers move at once — fuel prices, demand growth, and project lead times on the generation and on the transmission side. Every run covers 2025 to 2050 at a monthly step with an hourly dispatch inside each month.

The contribution is threefold:

1. **A quantified ranking of the uncertainties, not an illustration of them.** We report standardised one-at-a-time sensitivities and rank correlations over a Latin-hypercube ensemble, together with a convergence check on the reported percentiles. The ranking is unambiguous: demand growth dominates every indicator we track, and the other three drivers — including fuel prices, the one most often treated as the central risk for a gas-dependent system — are an order of magnitude weaker.
2. **A transmission-explicit representation of a longitudinal system.** Expansion is disaggregated into grid built for demand, grid built to connect new generation, and grid built to interconnect zones, each with its own trigger and its own lead time. This matters in Peru because the three categories respond to different drivers and only the third is constrained by the series topology.
3. **A reproducible artefact.** The model, the input workbook, the experimental design, the driver, the analysis code and a commented notebook that regenerates every figure are openly available. The

validation battery, the full parameter set and the extended results are given as supplementary material.

The central empirical result is counter-intuitive in a way that bears directly on policy. Higher demand growth does not raise the long-run average price in this system; it lowers it, because it pulls forward low-cost renewable capacity that the central case defers. The least-cost trajectory is consequently also the most transmission-intensive, requiring about a quarter more network than the central case. For a system whose grid is already the binding constraint, that is not a comfortable finding, and it reframes the planning problem: the risk to manage in Peru is not fuel-price exposure but whether transmission can be delivered fast enough to serve demand growth that is itself the dominant unknown.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature on transmission planning under energy transition, infrastructure delays, renewable integration in emerging economies, and system dynamics in energy planning, and positions the paper against it. Section 3 describes the Peruvian power system. Section 4 sets out the model, the experimental design and the validation protocol. Section 5 reports the central, best and worst cases and the uncertainty analysis. Section 6 discusses the implications and Section 7 concludes.

2. Literature review

2.1. Transmission planning under the energy transition

Transmission is the part of the power system where the energy transition is most visibly constrained. The resources that decarbonise a system are located where the wind blows and the sun shines, not where load is concentrated, so every increment of renewable capacity implies an increment of network; and the network is planned centrally, built slowly, and remunerated under rules written for a different generation fleet.

The modelling response has largely been optimisation. Transmission expansion planning (TEP) treats a central planner maximising social welfare subject to network constraints, and the literature has extended it steadily to accommodate the transition: stochastic formulations that make renewable hosting capacity an explicit objective [52], multi-stage investment games for offshore grids [49], and allocation models that coordinate renewable siting with inter-regional transfer capability [56]. Joint generation and transmission formulations (GTEP) go further, and the coordination gain they identify is substantial: Herding et al. [20] find that coordinating the two reduces total investment by around 10 % and network expansion by up to 30 % when distributed generation is allowed to displace transfers. Reviews of the field [18, 4] document both its maturity and its central assumption — a single rational agent optimising a well-posed objective.

That assumption is where the difficulty lies for the question we ask. In a liberalised market the generation investor and the network planner are different agents, acting on different information, over different horizons, and the literature itself notes that optimisation struggles to represent the structural change and the behavioural response that policy induces [12]. Recent work has begun to address the consequences rather than the optimum: how market design affects resilience under capacity expansion [33], how the representation of hydropower in capacity expansion models changes the result [30], and how the cost of capital — not resource quality or levelised cost — determines which Latin American projects actually get financed [11]. The last of these is directly relevant to Peru: it argues that the binding constraint in the region is institutional and financial rather than physical.

2.2. Infrastructure delays and power system reliability

A transmission project that is planned but late is, for the period of the delay, indistinguishable from a project that was never planned. The literature on this is smaller than the TEP literature but consistent in its

findings. Ritter et al. [37] simulate delayed interconnector expansion in a high-renewable European system and show that the delay does not merely postpone benefits: it raises emissions and generation costs, because surplus renewable energy that cannot be moved is replaced by conventional generation elsewhere, altering the final composition of the mix. Herrera et al. [21] reach a structurally similar conclusion for northeast Brazil, where transmission constraints are shown to bound wind expansion directly. The mechanism in both cases is the same: generation investment responds to a connection opportunity that the network has not yet delivered.

The causes of delay are well documented and largely non-technical. Right-of-way disputes, licensing, and local opposition recur across jurisdictions, and Sun et al. [48] quantify the last of these, showing that mitigating local opposition materially changes the feasible expansion path. Boateng et al. [6] generalise the point, identifying institutional-quality thresholds below which renewable deployment responds very differently to the same incentives. For a system such as Peru’s, where the auction mechanism that structured renewable procurement has been suspended, these findings suggest that lead-time risk is better modelled as an institutional variable than as an engineering one.

What is less settled is how much delay actually matters relative to the other things a planner does not know. Studies typically impose a delay scenario and report its consequences, which establishes that the effect exists without establishing its rank. That is the question this paper addresses.

2.3. Renewable integration and grid expansion in emerging economies

Emerging economies face the integration problem with a different constraint set: capital is more expensive, institutions are thinner, and the grid is often radial or longitudinal rather than meshed. da Silva and Carneiro [44] analyse the regional efficiency of renewable expansion in Brazil and find performance varying sharply by region, which is consistent with network access rather than resource quality being the differentiator. Filippo Antonioli et al. [15] document the role of residential rooftop prosumers in Brazil, a mechanism that reduces the inter-zonal transfers a longitudinal grid must support, and Wu and Zhong [54] reach a similar conclusion for community-scale rooftop solar and wind. Sheng et al. [43] make the link explicit by planning urban rooftop PV specifically to relieve grid-congestion-induced curtailment.

The alternative to building network is to use the existing network harder. Grid-enhancing technologies — dynamic line rating, power-flow control, advanced monitoring — raise effective transfer capability without new right-of-way, and Sadeghian and Mohammadi-Ivatloo [39] formalise dynamic thermal line rating under chance constraints for precisely this purpose. Their appeal in a system such as Peru’s is obvious, since the constraint is corridor capacity rather than generation adequacy; their limitation is that remuneration frameworks for them are generally absent, so deployment lags technical feasibility.

Storage plays the complementary role. Sousa et al. [46] quantify the substitution between storage and other flexibility resources in the Portuguese system, and Liu et al. [25] integrate multi-time-scale uncertainty and low-carbon retrofit into a regional capacity expansion scheme. Both make the same structural point: storage and network are partial substitutes, so the amount of transmission a system needs is not independent of how much flexibility it builds.

2.4. System dynamics in energy planning

System dynamics takes the opposite methodological stance to optimisation. Rather than solving for the best configuration, it simulates the configuration that the system’s own feedback structure, accumulations and delays will produce [17, 47]. For electricity this has proved productive because the pathologies of interest — investment cycles, capacity gaps, price spikes following construction lags — are consequences of structure rather than of optimisation failure.

The lineage is long. Ford [16] explained the California capacity crisis as a construction-delay phenomenon; Arango [2] simulated alternative regulation in the Colombian market; Assili et al. [3] examined capacity payments under long-term planning uncertainty, explicitly incorporating construction delays and demand growth; and Simshauser [45] connected resource adequacy to investment uncertainty in a liberalised market. More recent work has extended the approach to full transition pathways: Zapata et al. [55] analysed the long-run effects of a 100 % renewable target on the Colombian market, Morcillo et al. [28] examined demand-growth scenarios, and Luo et al. [26] built a hybrid system dynamics model of power-mix trajectories under carbon and capacity policy. Lin and Zhao [24] apply the method to the carbon border adjustment mechanism, and Oyarzún et al. [32] assess residential PV adoption policy with a system dynamics model — an indication that the approach remains current in this journal’s literature.

Model confidence in system dynamics is established by a documented battery of tests rather than by a goodness-of-fit statistic [5, 31], and recent methodological work has reconsidered what confidence means for models used exploratorily rather than predictively [36]. We follow that protocol explicitly in Section 4 and report it in full in the supplementary material, because a model used to rank uncertainties must first be shown to be structurally sound.

Where the system dynamics energy literature is thinnest is in uncertainty quantification. Scenario analysis is near-universal; designed sampling of the input space is rare. Outside system dynamics the tools are standard — Latin hypercube sampling [27] and variance-based or regression-based global sensitivity measures [22] — and they are used in power system work, for instance in Álvarez-Piñeiro et al.’s [2026] probabilistic assessment of the Spanish 2030 system by Monte Carlo analysis. Combining them with a feedback-rich simulation model is the methodological step this paper takes.

2.5. *Latin American and Peruvian electricity system studies*

Peru’s restructuring has been studied as a reference case. Rudnick and Velasquez [38] document a reform that delivered competition, lower wholesale prices and improved security of supply, while noting the accumulation of a high reserve margin and increasing fossil dependence; Pérez-Reyes and Tovar [34] measure the productivity gains in distribution following the reforms, which were real but concentrated. More recent work has turned to the transition itself. Fiestas-Chevez et al. [14] quantify the system-level impact of raising non-conventional renewables in the Peruvian matrix, finding reduced gas consumption, avoided emissions and lower system costs, and De la Cruz Torres et al. [10] reach compatible conclusions in long-range demand and emissions scenarios to 2050. On the network side, Valdivia et al. [51] report the congestion-analysis practice now used in Peruvian transmission planning, including $n - 1$ contingency screening with demand, generation, fuel-price and weather projections.

At regional scale, Sauma and Oren [40] established the proactive-versus-reactive distinction in transmission valuation that frames much of the subsequent debate, and analyses of Andean interconnection have long found the links technically and economically feasible. The obstacle identified is political rather than engineering: regulatory sovereignty, not transfer capability [19, 42]. Peru’s position in the regional comparison is genuinely ambiguous — early policy innovation through auctions, substantial resources, and yet no consolidated decarbonisation strategy [8, 35, 9, 50].

2.6. *Research gap and positioning*

Three gaps follow from the above.

First, the delay literature establishes that lead times matter but not how much they matter relative to other uncertainties. Delay is typically imposed as a scenario; it is rarely ranked against demand, fuel prices or cost assumptions in a common design.

Second, the system dynamics energy literature rarely quantifies uncertainty with a designed experiment. The strength of the method is that it reproduces counter-intuitive behaviour arising from structure; the corresponding weakness, when it is used for policy, is that a small number of named scenarios cannot separate a structural result from an artefact of one input assumption.

Third, Peru is under-represented. The existing Peruvian studies are either optimisation-based assessments of a renewable target [14, 10] or operational analyses of the network [51]. None represents the investment-and-delay feedback between generation and transmission that a longitudinal, gas-dependent system with suspended auctions would be expected to exhibit.

This paper addresses all three: a system dynamics model of the Peruvian system with explicit generation and transmission pipelines, subjected to a Latin-hypercube designed experiment that ranks four uncertainties against each other on a common footing, with the validation battery and the full input set reported.

3. The Peruvian power system

3.1. Overview and institutional structure

In the late 1980s Peru's power sector was in crisis, driven by a broader macroeconomic collapse and poorly performing utilities. The response was a structural reform: the country moved from an integrated state system to a competitive wholesale market with marginal nodal prices, capacity payments and bilateral contracts, which delivered efficiency gains, new installed capacity and a more secure supply [38]. The Electric Concessions Law of 1992 unbundled what had been a single chain into generation, transmission and distribution, opening the sector to private investment. A subsequent wave of reforms, including the supply auctions introduced from 2006, consolidated productivity gains, with clear efficiency improvements in distribution after restructuring [34]. Challenges persist in state-owned companies and in rural coverage.

3.2. Generation mix and dependence on thermal generation

The Peruvian generation matrix is dominated by hydropower and natural-gas-based thermal generation. By 2015 the major technologies were roughly 50 % hydro and 45 % natural gas, with thermal sources overall — gas, coal and oil — accounting for approximately 41 % of generation by 2017 [38]. This gas dependence is structural rather than incidental: over half of all natural gas consumed in the country goes to power plants [13], which makes the electricity sector the primary driver of gas demand and leaves it exposed to price shocks and supply disruptions.

Despite substantial potential in solar, wind, geothermal, biomass and hydroelectric resources, non-hydro renewables have remained marginal. Wind, solar and biomass together contributed only 2.8 % of net generation in 2017 [38], and by 2022 only around 5 % of electricity generation came from non-conventional renewable sources [8]. There has been diversification into solar photovoltaics, wind, biomass and small hydro, but thermal dependence on natural gas remains the defining structural feature of the matrix.

3.3. Renewable expansion potential

Peru's resource base is substantial — solar radiation, coastal and highland wind, geothermal activity, biomass and hydroelectric potential are all present in significant measure — yet little of it has been converted into generation capacity. Fiestas-Chevez et al. [14] find that raising non-conventional renewables to 28.26 % of the matrix would cut gas consumption by 2.04 km³ and avoid 4.99 Mt CO₂-eq annually while reducing overall system costs. Longer-range scenarios to 2050 point the same way: prioritising renewable integration alongside transmission and distribution efficiency improvements produces significantly lower emissions than the baseline trajectory [10]. The economic and environmental case is, in other words, reasonably well established.

Table 1: Initial condition of the Peruvian system, January 2025, as represented in the model. Capacity is the sum over the technologies present in each zone; corridor capacity is the initial inter-zonal transfer limit.

Zone	Description	Installed capacity [MW]	Share
Z1	North	1,606.1	11.4 %
Z2	Centre	8,461.9	60.2 %
Z3	South	3,989.0	28.4 %
Total		14,057.0	100 %
<i>Interconnection corridors</i>			
Z1–Z2	232.3 km corridor	900 MW	
Z2–Z3	271.8 km corridor	700 MW	
Total transmission network		16,180 km	

What is less clear is how to get there. The auction mechanism that for over a decade provided a structured channel for renewable investment has been suspended, leaving developers without a reliable off-take signal [8]. The surrounding conditions make this harder: excess installed capacity depresses wholesale prices, economic growth has slowed, and the regulatory framework is fragmented across institutions with overlapping mandates. On top of that, the grid was not built with variable renewables in mind, and the mismatch between where the best resources are and where demand is concentrated remains an unresolved physical constraint [50]. The international evidence suggests this combination is not unusual: Boateng et al. [6] show that institutional quality conditions how strongly renewable deployment responds to otherwise identical incentives, and Del Río et al. [11] find that in Latin America the cost of capital, rather than resource quality or levelised cost, determines which projects are financed.

3.4. Transmission corridors and the zonal structure

Transmission infrastructure is both a current bottleneck and a strategic lever for Peru’s energy transition. Within the national system, congestion management has become more sophisticated, with $n - 1$ contingency screening alongside demand, generation, fuel-price and weather projections, enabling early detection of long-term congestion points and supporting a network that can better accommodate renewable sources while reducing operational costs [51].

The model represents the system as three zones connected in series, shown in Figure 1. This is not a simplification imposed for tractability but a representation of the physical topology: Peru’s main grid runs longitudinally, and the consequence is that the southern zone reaches the rest of the system only through the centre. Table 1 reports the January 2025 starting condition. The centre zone (Z2) holds 60 % of installed capacity, the two corridors Z1–Z2 and Z2–Z3 provide 900 MW and 700 MW of transfer capability respectively, and there is no direct Z1–Z3 path. A series topology of this kind has no redundancy: a constraint on the central corridor isolates a third of the system, and every inter-zonal transfer between the extremes consumes capacity on both corridors.

The initial technology mix, reported in Table 2, shows the structural features described above: thermal plant and reservoir hydro together account for 78 % of capacity, while wind and solar combined contribute 10.7 %.

At the regional level, analyses of Andean interconnection indicate that technically and economically feasible links among Bolivia, Chile, Colombia, Ecuador and Peru would generate long-run economic and environmental benefits. These opportunities remain largely unexecuted. Unlike Central America, which has advanced toward a regional electricity market through supranational institutions, the Andean region has



Figure 1: Zonal representation of the Peruvian power system. Three zones are connected in series: north (Z1), centre (Z2) and south (Z3). There is no direct Z1–Z3 corridor, so all exchange between the extremes transits the centre.

stalled, primarily because member states are reluctant to cede regulatory sovereignty, despite transmission infrastructure that makes integration technically viable [42, 19]. The gap between technical feasibility and political will, rather than any engineering constraint, is the principal barrier.

3.5. Planning challenges and infrastructure delays

The gains from Peru’s 1990s reform were real but uneven, and over time the limits of the institutional design became apparent. Once the initial wave of privatisation ran its course, new investment in generation and transmission slowed considerably, a sign that the regulatory framework was not generating the long-term certainty that infrastructure capital requires [38]. State-owned distributors have continued to underperform, rural electrification remains incomplete, and government has periodically intervened outside market mechanisms in ways that distort price signals.

Table 2: Installed generation capacity by technology, January 2025.

Technology	Capacity [MW]	Share
Thermal (gas, group 2)	4,630.6	32.9 %
Hydro — reservoir	3,705.6	26.4 %
Thermal (gas, group 1)	2,622.7	18.7 %
Hydro — run-of-river	1,532.5	10.9 %
Wind — onshore	1,021.3	7.3 %
Solar PV	477.8	3.4 %
Biomass	66.6	0.5 %
Total	14,057.0	100 %

These shortcomings matter now, as the country attempts to expand its renewable base. The suspension of the auction mechanism has hit onshore wind hardest: projects conceived for a structured procurement environment are instead being asked to compete in a spot market under considerable price uncertainty [50]. Beyond procurement, wind development faces a geographic difficulty — the best resources are far from consumption, and the grid lacks the flexibility to bridge that gap. Taken together, fossil-fuel dependence and a renewable share of roughly 5.5 % of total energy leave Peru exposed to external price swings while limiting access to cleaner supply, a situation that will not improve without more deliberate institutional backing [9].

This is the feature of the Peruvian case that motivates the modelling choice in Section 4. Lead times here are not an engineering parameter but an institutional outcome, and they differ systematically between generation, which responds to a market signal, and transmission, which is planned centrally and delivered on a longer clock. Representing that asymmetry explicitly is what allows the model to be asked which of the two actually binds.

3.6. Relevance of the Peruvian case

Peru occupies a distinctive position in comparative analyses of Latin American energy transitions, functioning simultaneously as a reference case for successful market reform and as a cautionary example of its limits. The post-1992 restructuring achieved genuine competition, lower wholesale prices and improved security of supply, but at the cost of increasing fossil-fuel dependence and a high reserve margin that raises questions about long-term allocative efficiency [38]. Peru was also among the regional pioneers of technology-specific renewable auctions, which produced price reductions of 78 % for solar and 53 % for wind and demonstrated that competitive procurement can scale non-conventional renewables in emerging markets.

Early policy innovation has not, however, translated into a consolidated transition strategy. Legal reforms have accumulated incrementally rather than forming a unified framework, and no comprehensive national decarbonisation plan exists — a gap visible when Peru is set against Chile, which started from a comparable liberalised base and has moved considerably further [35]. Peru’s situation reflects a tension that runs through the region: the same country holds major hydrocarbon reserves, Camisea being the most significant, alongside world-class solar, wind, biomass and green hydrogen potential. How governments navigate that duality, and the related choices about gas as a transitional fuel, hydropower expansion and regional grid integration, will do much to determine what Latin America’s energy system looks like in the long run [19].

4. Methodology

4.1. Modelling approach

The question posed in Section 1 — which uncertainty governs the trajectory of the Peruvian system — requires two things that are usually provided separately. It requires a model whose behaviour arises from the structure of the decision process rather than from an optimality assumption, because the phenomenon of interest is the mismatch between a decentralised generation market and a centrally planned network. And it requires a designed experiment over the inputs, because a ranking cannot be read off a handful of named scenarios.

We therefore combine a system dynamics simulation model with a Latin-hypercube designed experiment. The simulation runs monthly from January 2025 to January 2050 (300 months, integration step 0.25 month), with an hourly dispatch represented inside each month by one representative day of 24 hourly

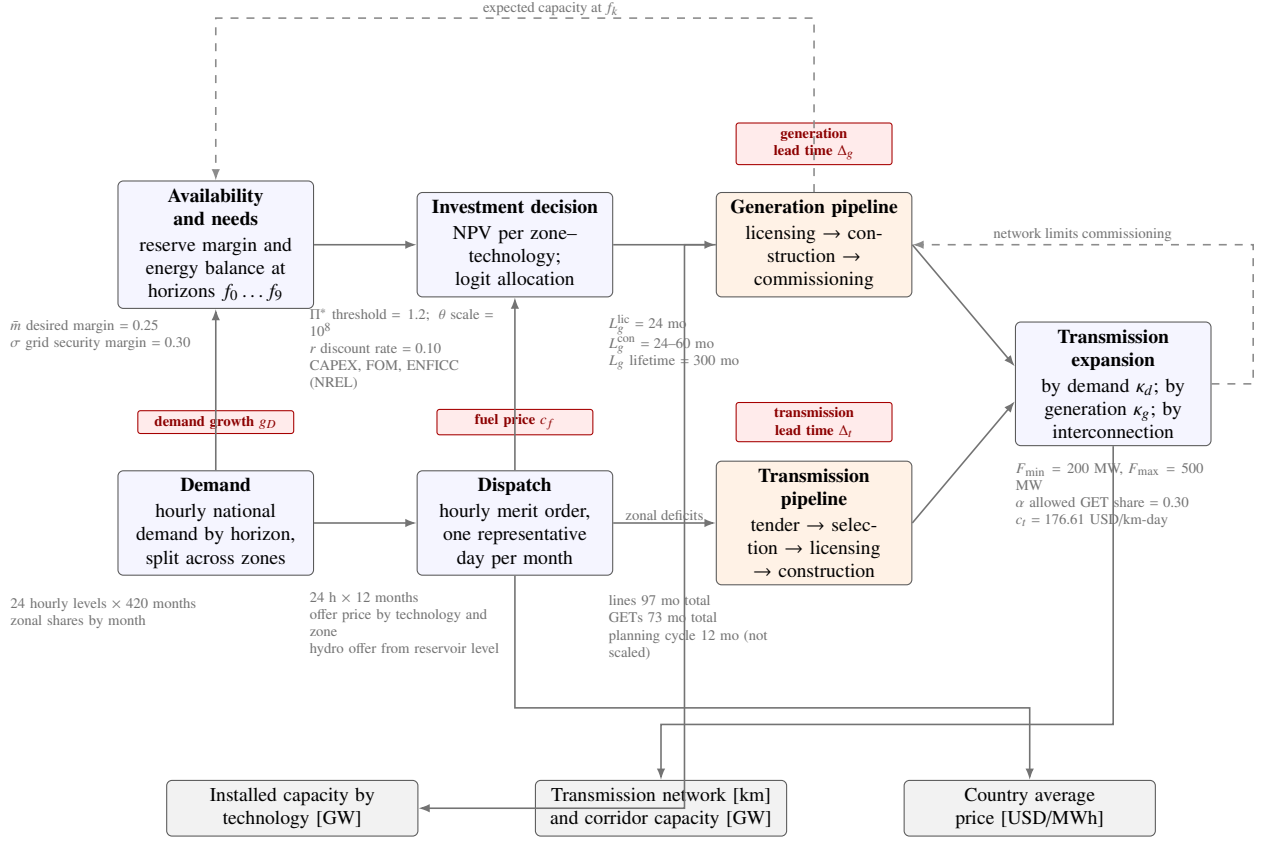


Figure 2: Structure of the model. Solid arrows are material and information flows within a simulation step; dashed arrows are the two feedbacks that generate the behaviour of interest — commissioning is constrained by the network that has actually been delivered, and investment decisions are taken on expected rather than current capacity. Grey text gives the governing parameters and their central values; the four boxes in red are the uncertain drivers varied in the designed experiment of Section 4.3.

levels. The experiment comprises 129 runs: one central case, two one-at-a-time runs per factor for the tornado diagram (eight), and 120 Monte Carlo samples in which all four factors vary simultaneously.

The division of labour between the two is deliberate. The one-at-a-time runs isolate each factor against a fixed reference, which is what makes a tornado bar attributable to that factor and nothing else; their limitation is that they say nothing about interaction. The Monte Carlo ensemble moves everything at once, which is what the percentile envelopes and the rank correlations require; its limitation is that an individual factor's effect is no longer separable by construction and must be inferred statistically. Reporting both, and checking that they agree on the ranking, is stronger than either alone.

4.2. Model structure

Figure 2 shows the modules, the flows between them, and the parameters governing each. The notation used throughout is collected in Table 3.

4.2.1. Generation: investment, pipeline and commissioning

Generation capacity is a stock fed by a pipeline of projects that must pass through licensing and construction before they can be commissioned. Writing $N_{z,g}(t)$ for the capacity committed in month t :

Table 3: Notation.

Symbol	Meaning
$z \in \mathcal{Z}$	zone, $ \mathcal{Z} = 3$ active of 6 represented
$g \in \mathcal{G}$	generation technology, $ \mathcal{G} = 14$ per zone
$i \in \mathcal{I}$	interconnection corridor, $ \mathcal{I} = 2$ active of 15
$h \in \mathcal{H}$	hour of the representative day, $ \mathcal{H} = 24$
$f \in \mathcal{F}$	forecast horizon, $\mathcal{F} = \{f_0, f_1, f_3, f_5, f_7, f_9\}$ years ahead
t	time, months from January 2025
$K_{z,g}(t)$	installed generation capacity [MW]
$Q_{z,g}^{\text{lic}}(t), Q_{z,g}^{\text{con}}(t)$	capacity in licensing / under construction [MW]
$T_z^d(t), T_z^g(t), T^i(t)$	transmission for demand / generation / interconnection [km]
$P_i(t)$	transfer capability of corridor i [MW]
$D_{z,h,f}(t)$	demand [MW]
$\lambda_{z,h}(t)$	zonal price [USD MW ⁻¹ h]
$\bar{m}$	desired availability margin, 0.25
σ	grid security margin, 0.30
Π^*	profitability threshold, 1.2
θ	logit scale for technology allocation, 10 ⁸ USD
r	discount rate, 0.10
κ_d, κ_g	grid intensity of demand and generation [km MW ⁻¹]
$F_{\min}, F_{\max}$	minimum / maximum grid increment, 200 / 500 MW
α	share of a corridor's need that a GET may serve, 0.30
c_t	transmission cost, 176.61 USD km ⁻¹ -day
Δ_g, Δ_t	lead-time factors, generation and transmission
c_f, g_D	fuel-price and demand-growth factors

$$\frac{dQ_{z,g}^{\text{lic}}}{dt} = N_{z,g}(t) - \mathcal{D}[N_{z,g}, L_{z,g}^{\text{lic}}](t), \quad (1)$$

$$\frac{dQ_{z,g}^{\text{con}}}{dt} = \mathcal{D}[N_{z,g}, L_{z,g}^{\text{lic}}](t) - \mathcal{D}[N_{z,g}, L_{z,g}^{\text{lic}} + L_{z,g}^{\text{con}}](t), \quad (2)$$

$$\frac{dK_{z,g}}{dt} = \min\{\mathcal{D}[N_{z,g}, L_{z,g}^{\text{tot}}](t), \Phi_z(t)\}, \quad (3)$$

where $\mathcal{D}[\cdot, L]$ is a material delay of duration L and $L_{z,g}^{\text{tot}} = (L_{z,g}^{\text{lic}} + L_{z,g}^{\text{con}})(1 + \Delta_g)$ is the total lead time including the lead-time factor. The term $\Phi_z(t)$ in (3) is the headroom the delivered network allows in zone z : capacity that has finished construction cannot be commissioned until there is transmission to evacuate it. Equation (3) is the first of the two feedbacks drawn dashed in Figure 2, and it is what couples the generation market to the network plan.

Investment $N_{z,g}(t)$ is triggered by two distinct mechanisms. The first is adequacy: when the expected reserve margin at horizon f falls below the desired margin $\bar{m}$, the shortfall is converted into a capacity requirement. The second is profitability: for each zone–technology pair the model computes a net present value over a 25-year economic life, assuming the plant is the only entrant and that current plus already-committed capacity persists, using CAPEX, fixed O&M, fuel and firm-capacity (ENFICC) parameters from National Renewable Energy Laboratory [29]. Options whose profitability index exceeds Π^* are admitted.

Admitted options are then allocated across technologies by a logit share with scale θ , which represents the dispersion of investor beliefs rather than a single optimising agent — the point at which the model departs most clearly from a GEP formulation.

Decisions are taken on *expected* rather than current state. Expected capacity at horizons of 3, 5, 7 and 9 years includes capacity already in the pipeline, so investors and planners are assumed to know the construction times they face. This is the second dashed feedback in Figure 2 and it is what prevents the model from over-investing in response to a deficit that is already being remedied.

4.2.2. *Transmission: three categories, three triggers*

Transmission expansion is disaggregated into three categories because in this system they respond to different drivers and carry different lead times.

Transmission for demand. Peak demand growth in a zone requires network inside that zone, at an empirical intensity κ_d [km MW⁻¹]: $T_z^d(t) = \kappa_d \hat{D}_z(t)$, with $\hat{D}_z$ the zonal peak. This category carries no construction delay, because the model does not represent intra-zonal networks and a delay there would not affect inter-zonal operation.

Transmission for generation. Each new plant requires the network without which it could not reach commercial operation, at intensity κ_g . The trigger is the formal identification of a project at a planning horizon, which makes this category structurally *reactive*: it responds to confirmed projects rather than anticipating them, and because transmission lead times exceed generation lead times, this is a principal source of the bottleneck the model is built to study.

Transmission for interconnection. Corridor reinforcement is driven by the dispatch. At each step the dispatch identifies zonal surpluses and deficits and the implied inter-zonal transfers; where the required transfer exceeds the delivered corridor capability $P_i(t)$, a reinforcement is triggered. If the shortfall is at most a share α of current capability it may be met by a grid-enhancing technology, which has a shorter pipeline; otherwise a new line is required. Increments are bounded below and above by $F_{\min}$ and $F_{\max}$, which represents the lumpiness of real reinforcements.

Transmission projects pass through a four-stage pipeline — public tender, investor selection, licensing, construction — each stage scaled by $(1 + \Delta_i)$. The 12-month planning cycle is deliberately *not* scaled: the model uses it as the period of a modulo operation that sets when plans are revised, so scaling it would shift the planning calendar rather than delay projects.

4.2.3. *Dispatch and prices*

Each month the model dispatches one representative day of 24 hourly levels by merit order. Each zone–technology pair is represented by one plant with an offer price and an hourly availability profile; reservoir hydro offers between a minimum and a scarcity price as a function of its storage level, which is what gives hydrology a price signal rather than merely an energy constraint. This is a reduced-form representation of reservoir operation: a full treatment would optimise releases against inflow forecasts and environmental constraints [41], which matters more for operational studies than for the expansion question asked here. Two dispatches are computed: an unconstrained one, used to size the transfers the network would ideally carry and hence to detect reinforcement needs, and a constrained one that respects delivered corridor capability and yields congestion, zonal prices and the realised generation mix.

The delivered price reported in this paper is the country average electricity tariff, obtained from the zonal tariffs over the active zones. Because the price oscillates month to month with hydrology, the reported indicator is the mean over the final twelve months of the horizon rather than a single month; a single month is not representative of the state the system has reached.

Table 4: Uncertain factors, ranges, and the model quantity each one scales. Ranges are ranges of admissibility, not estimated distributions.

Factor	Range	Model quantity
Demand growth g_D	$-20 / +20 \%$	multiplier on national demand, equal to 1 in January 2025 and $1+g_D$ in January 2050, so only the growth path changes
Generation lead time Δ_g	$-20 / +20 \%$	licensing plus construction time of every generation technology
Transmission lead time Δ_t	$-20 / +20 \%$	tender, investor selection, licensing and construction of lines and GETs
Fuel price c_f	$-15 / +15 \%$	fuel cost of every technology that burns fuel

The model also supports an external dispatch path in which the constrained dispatch is solved as a linear program with PyPSA [7] through the Vensim DLL. All results reported here use the internal rule-based dispatch, for consistency with the central case against which the ensemble is compared.

4.2.4. Distributed resources and storage

Distributed PV adoption is represented as a diffusion process driven by the gap between the levelised cost of distributed generation and the retail tariff, moderated by a potential market and a social-influence term — the structure used in the system dynamics adoption literature [32] and consistent with the empirical role documented for prosumers in neighbouring systems [15, 54]. Battery storage, where enabled, arbitrages between the cheapest and the most expensive hour of the representative day. Both reduce the residual demand seen by the dispatch and hence, indirectly, the inter-zonal transfers the network must support — the substitution between storage and network quantified by Sousa et al. [46].

In the configuration used for every run reported here, system-level storage and grid-enhancing technologies are *disabled* at the switch level, as in the shipped baseline, while distributed resources remain endogenous. PV-plus-storage *technologies* can still be selected by the investment rule and do appear in the results. This is stated explicitly because it bounds the interpretation: the study quantifies uncertainty around the current policy configuration, not the effect of deploying storage or GETs.

4.3. Experimental design

4.3.1. Factors and ranges

Four drivers are varied. They were selected because each is uncertain, each is outside the planner’s direct control on the relevant horizon, and each has been argued in the literature to be a principal risk for a system with Peru’s characteristics.

Two conventions require comment. First, the demand factor is a *growth-path* factor, not a level shift: the multiplier is unity at the start of the horizon and reaches $1 + g_D$ at the end, so January 2025 demand is untouched and only the rate of growth varies. This is the correct construction for an expansion study, where the quantity in question is how fast load grows, not how large it is today. Second, generation lead times are expressed relative to the *workbook base case* rather than to the planned time. The Peruvian data set carries an “other delays” fraction of -0.2 , meaning realised delay is 80 % of planned; the factor is therefore applied as $(1 + \text{base})(1 + \text{change}) - 1$, giving -0.36 at the low end and -0.04 at the high end. This is the same

convention used for transmission, which is what makes the two delay bars of the tornado comparable with each other.

Distributions are uniform and independent. There is no information that would justify another shape, and we are explicit that these are ranges of admissibility rather than estimated distributions: the ranking is robust to this choice, the absolute width of the percentile band is not.

4.3.2. Sampling

The Monte Carlo sample is a Latin hypercube [27]: each factor's range is divided into as many equal intervals as there are samples, one value is drawn at random inside each interval, and the intervals of the factors are paired at random. This gives even marginal coverage with far fewer runs than simple random sampling, which matters when a single run costs several minutes of simulation. The design is generated from a fixed seed and written to disk before any simulation starts, so the experiment is fully specified independently of its execution.

The realised design has a maximum absolute correlation between factors of 0.181 and a mean of 0.076. Near-orthogonality matters because the rank correlations reported in Section 5.2 are only attributable to a single factor when the factors are close to independent.

4.3.3. Central, best and worst case

We report three trajectories in detail, and the basis for choosing them is stated rather than left implicit.

The **central case** is the system run with every factor at its reference value — the configuration of the input data set as compiled from Peruvian sources. It is not a forecast and it is not the ensemble median; it is the reference against which every other run is measured, and it is reported because the tornado diagram's bars are defined as departures from it.

The **best** and **worst** cases are selected from the 120 Monte Carlo runs on a single declared metric: the lowest and the highest country average electricity price, averaged over the final twelve months. A single declared criterion keeps the selection auditable, which is the point of reporting two runs in full; a weighted composite over several indicators would make the choice depend on weights that have no defensible basis. The consequence — that “best” on price need not be best on every other indicator — is not a defect of the criterion but a result, and we return to it in Section 5.4.

4.4. Implementation and validation

The model is implemented in Vensim DSS [53] and executed through its DLL from a Python driver that applies each sample, runs the simulation, and extracts the reported variables. Each uncertain factor enters the model as a named constant multiplying the corresponding structural equation, with a default value that reproduces the unmodified model exactly. The complete driver, the design and the analysis code are available with the paper.

Model confidence is established by the standard system dynamics battery [5, 31, 36] rather than by a single fit statistic, which is the appropriate standard for a model used to explore structure rather than to forecast. The tests are reported in full in Supplementary Material S3; in summary:

- *Structure verification.* Every feedback loop is matched to a documented decision process and a source.
- *Parameter verification.* Every parameter is reported with its value, unit, source and admissible range (Supplementary Material S2).
- *Dimensional consistency.* The model passes Vensim's units check.

- *Extreme-condition tests.* Six tests, including zero demand growth, zero construction time and unbounded lead times, each with the behaviour that structural soundness requires.
- *Behaviour reproduction.* The central case reproduces the documented reference trajectory of the Peruvian system, and the published four-factor model reproduces the central case to 63.85 GW and 46,341.23 km at January 2050, matching the reference exactly.
- *Behaviour sensitivity.* The designed experiment of Section 4.3 is itself the sensitivity test, and the convergence of the reported percentiles is checked in Section 5.2.

5. Results

5.1. Central case: the trajectory the reference data imply

Under the reference values of every factor, the Peruvian system roughly quadruples its installed generation capacity between 2025 and 2050, from 14.06 GW to 63.85 GW, while the transmission network grows from 16,180 km to 46,341 km and inter-zonal transfer capability from 1.60 GW to 4.56 GW. The country average electricity price falls from roughly 210 USD MW⁻¹ h in the late 2020s to 124.5 USD MW⁻¹ h by 2050.

The composition of that growth is the first result worth noting. Almost all of it is onshore wind and solar: wind rises from 1.02 GW to 34.83 GW and solar PV from 0.48 GW to 8.33 GW, while thermal capacity grows only from 7.25 GW to 9.65 GW. By 2050 non-thermal technologies hold 84.9 % of capacity, against 48.4 % at the start. The model reaches this outcome without any renewable target, mandate or carbon price: expansion is driven by the profitability rule alone. Thermal plant is not retired — the model has no endogenous retirement flow — but it stops being the marginal investment.

The price path is not monotone. It rises through the late 2020s, when demand grows against a capacity base that cannot yet respond, and falls thereafter as wind and solar enter. That shape is the classic capacity-cycle signature that system dynamics models of liberalised markets reproduce [16, 2, 45], and it appears here without being imposed.

5.2. Which uncertainty governs the system

Figure 3 reports the one-at-a-time sensitivity of each indicator to each factor. The result is unambiguous. Demand growth dominates every indicator by roughly an order of magnitude over the next-ranked factor, and fuel prices — the risk most often emphasised for a gas-dependent system — are the weakest driver of three of the four indicators.

Table 5 gives the numbers. A $\pm 20\%$ band on the 2050 demand level moves required transmission from -18.9% to $+23.8\%$ and installed capacity from -24.0% to $+29.3\%$; the swing attributable to demand is nine times that of generation lead times for transmission length, and five times for installed capacity. Fuel prices move installed capacity by at most 1.4 % and the average price by 0.4 %, in both directions — the latter is not a sign error but a genuine non-monotonicity, discussed below.

The one-at-a-time result is confirmed when all four factors move together. Table 6 reports Spearman rank correlations over the 120 Monte Carlo runs. Demand growth reaches 0.985 for transmission length and 0.973 for installed capacity — close to the maximum a rank correlation can attain — while no other factor exceeds 0.27 in absolute value on any indicator. The two methods agree on the ranking, which is the check that matters: a one-at-a-time result that survived only in the absence of interaction would not support the conclusion.

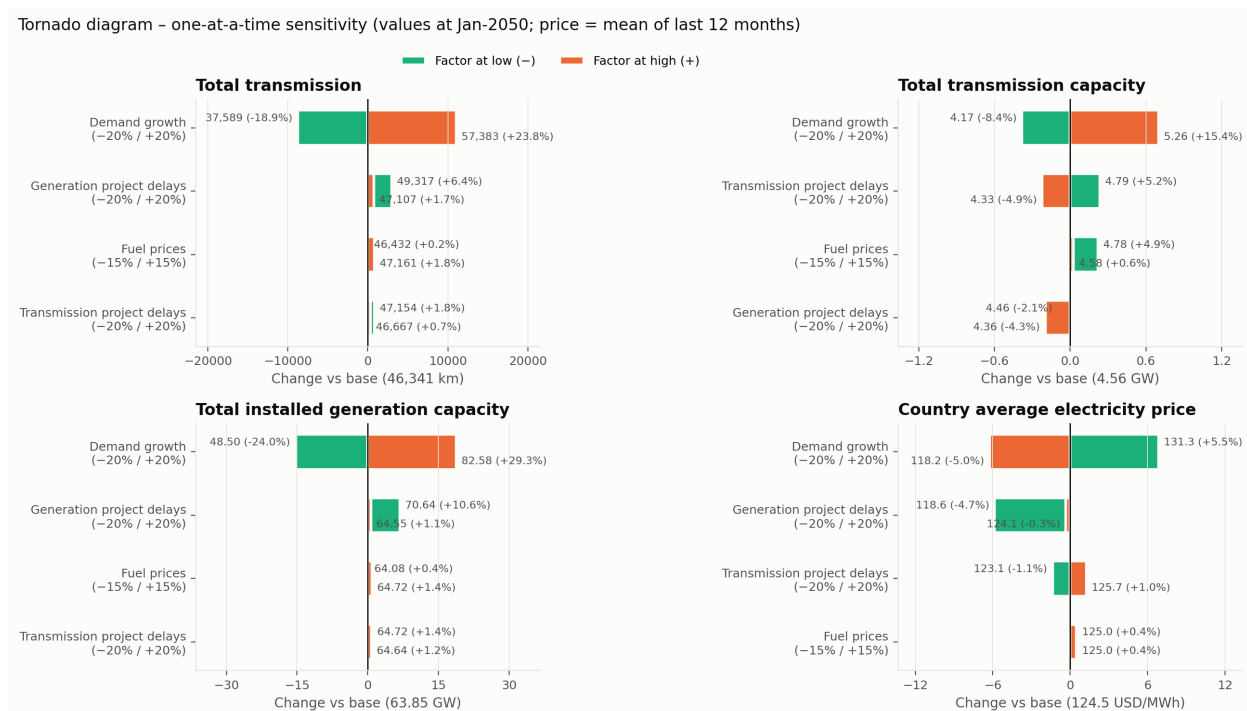


Figure 3: Tornado diagram. For each indicator, the bars show the change from the central case when one factor is set to the low and the high end of its range with the other three held at their reference value. Factors are ordered by swing, $|\text{high} - \text{low}|$. Values at January 2050; the price is the mean of the final twelve months.

5.3. How wide the range of outcomes remains

Figure 4 shows the P10–P90 envelope of each indicator over the whole horizon. The envelopes are narrow until the early 2030s and widen steadily thereafter, which is the expected signature of a system whose divergence is driven by a growth rate rather than by a level: small differences in the rate compound.

By 2050 the ranges are substantial, as Table 7 reports: installed capacity spans 52.3 GW to 82.5 GW between the tenth and ninetieth percentile, a factor of 1.6, and required transmission spans 39,858 km to 57,069 km. The price range is comparatively narrow, 116.8 USD MW^{–1} h to 129.4 USD MW^{–1} h, a spread of about 10 % around the median. The asymmetry between the two is itself informative: what these uncertainties move is *how much infrastructure the country has to build*, far more than *what electricity ends up costing*.

The central case sits close to the median on transmission and capacity but at the upper end of the price distribution. That is a consequence of the price non-monotonicity identified in Table 5: because both an increase and a decrease in several factors lower the price relative to the reference, the reference is near a local maximum.

Supplementary Material S4.2 reports the convergence of these percentiles as runs are added. P10, P50 and P90 are stable from roughly 75 runs onwards for every indicator, which is the evidence that 120 samples are sufficient rather than an assumption that they are.

5.4. Central, best and worst case

Table 8 reports the three trajectories selected as described in Section 4.3.3. The best case on delivered price is run mc_009, in which demand grows 17.3 % faster than the reference; the worst is mc_058, in which

Table 5: One-at-a-time sensitivity. Change from the central case at January 2050 with one factor at each end of its range. Factors ordered by swing within each indicator.

Indicator	Factor	Low	High	Swing
Transmission km, base 46,341	Demand growth	−18.9%	+23.8%	19,794
	Generation lead time	+6.4%	+1.7%	2,210
	Fuel price	+0.2%	+1.8%	729
	Transmission lead time	+1.8%	+0.7%	487
Installed capacity GW, base 63.85	Demand growth	−24.0%	+29.3%	34.08
	Generation lead time	+10.6%	+1.1%	6.09
	Fuel price	+0.4%	+1.4%	0.64
	Transmission lead time	+1.4%	+1.2%	0.08
Transfer capability GW, base 4.56	Demand growth	−8.4%	+15.4%	1.08
	Transmission lead time	+5.2%	−4.9%	0.46
	Fuel price	+4.9%	+0.6%	0.19
	Generation lead time	−2.1%	−4.3%	0.10
Average price USD MW ^{−1} h, base 124.5	Demand growth	+5.5%	−5.0%	13.09
	Generation lead time	−4.7%	−0.3%	5.52
	Transmission lead time	−1.1%	+1.0%	2.63
	Fuel price	+0.4%	+0.4%	0.01

Table 6: Spearman rank correlation between each factor and each indicator over the 120 Monte Carlo runs, in which all four factors vary simultaneously.

Factor	Transmission [km]	Transfer [GW]	Capacity [GW]	Price [USD MW ^{−1} h]
Demand growth	0.985	0.835	0.973	−0.752
Transmission lead time	0.112	−0.142	0.121	−0.081
Generation lead time	0.044	0.081	0.053	−0.264
Fuel price	−0.008	−0.013	−0.014	−0.014

it grows 16.5 % slower.

This is the central empirical finding of the paper, and it runs against the intuition that higher demand means a more stressed and therefore more expensive system. In this model the opposite holds: faster demand growth *lowers* the long-run average price. The mechanism is visible in the technology mix. Figures 5 and 6 show installed capacity and annual generation by technology for the three cases. In the best case, onshore wind reaches 52.05 GW against 34.83 GW in the central case and 25.47 GW in the worst; solar PV reaches 11.51 GW against 8.33 and 5.24 GW. Faster demand growth sustains a price signal long enough for low-cost renewable capacity to clear the profitability threshold earlier and in greater volume; once that capacity is in place it depresses the marginal price. Slow demand growth, conversely, suppresses the investment signal, leaves the system leaning on existing thermal plant — thermal generation is 10.2 TW h yr^{−1} in

Table 7: Indicator values at January 2050 over the 120 Monte Carlo runs.

Indicator	Central	P10	P50	P90	P90–P10
Transmission [km]	46,341	39,858	47,788	57,069	17,211
Transfer capability [GW]	4.56	4.23	4.70	5.29	1.06
Installed capacity [GW]	63.85	52.33	66.82	82.49	30.16
Average price [USD MW ^{−1} h]	124.5	116.8	122.8	129.4	12.59

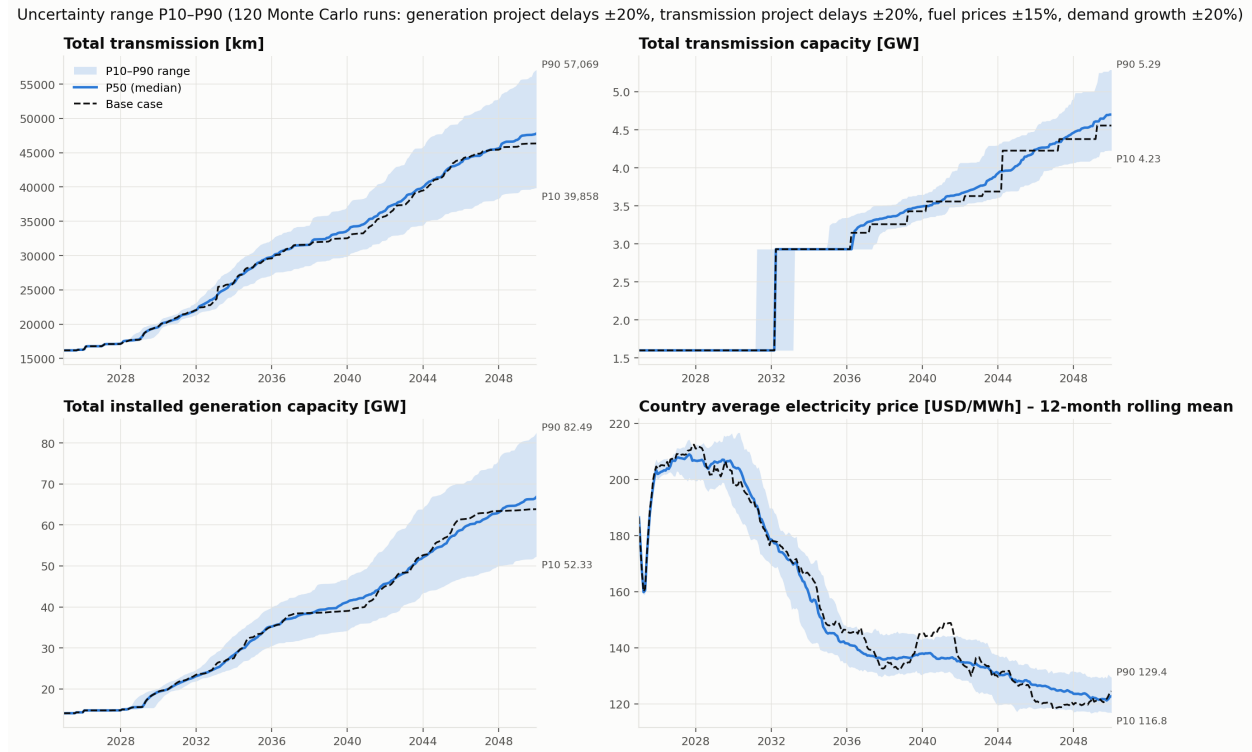


Figure 4: P10–P90 envelope of the four indicators over 120 Monte Carlo runs, with the ensemble median and the central case. The band contains the central 80 % of outcomes.

the worst case against 5.6 TW h yr^{-1} in the best — and ends with a higher average price despite a smaller system.

The corollary is uncomfortable for a planner. The least-cost trajectory is also the most infrastructure-intensive: the best case requires 58,562 km of network, 26.4 % more than the central case and 50 % more than the worst case, together with 10.3 % more inter-zonal transfer capability. The cheapest electricity in this system is not obtained by building less; it is obtained by building more, faster. Figure 7 shows the resulting price paths and Figure 8 the transmission requirement.

A second observation concerns the delay factors. The best case has *longer* generation lead times than the central case (+9.8 %) and the worst case has shorter ones (−5.4 %). This is not evidence that delays are beneficial; it is a consequence of their weakness relative to demand. Within the sampled ranges, the demand draw determines the outcome and the delay draw is close to noise, so the extreme runs on price carry essentially arbitrary delay values. Reporting this explicitly is more honest than selecting extreme runs that happen to tell a tidier story, and it is itself the clearest statement of the ranking: in this system, over these ranges, lead times do not decide the outcome.

5.5. Where the uncertainty acts

Decomposing transmission by category (Figure 8) locates the effect. Network built for demand connection scales directly with peak demand and therefore inherits the demand factor’s variance in full. Network built to connect generation follows the investment decision and inherits it indirectly. Inter-zonal corridor capability is the least sensitive of the three, and it is the only one where transmission lead time ranks second

Table 8: Central, best and worst case. Best and worst are the Monte Carlo runs with the lowest and the highest country average price over the final twelve months. Percentages are relative to the central case.

	Central	Best (mc_009)	Worst (mc_058)
<i>Factor values</i>			
Demand growth	0.0%	+17.3%	-16.5%
Generation lead time	0.0%	+9.8%	-5.4%
Transmission lead time	0.0%	+2.6%	+8.2%
Fuel price	0.0%	-2.5%	-1.1%
<i>Indicators at January 2050</i>			
Transmission [km]	46,341	58,562 (+26.4%)	38,927 (-16.0%)
Transfer capability [GW]	4.56	5.03 (+10.3%)	4.34 (-4.8%)
Installed capacity [GW]	63.85	86.40 (+35.3%)	49.18 (-23.0%)
Average price [USD MW ⁻¹ h]	124.5	114.9 (-7.6%)	135.6 (+9.0%)

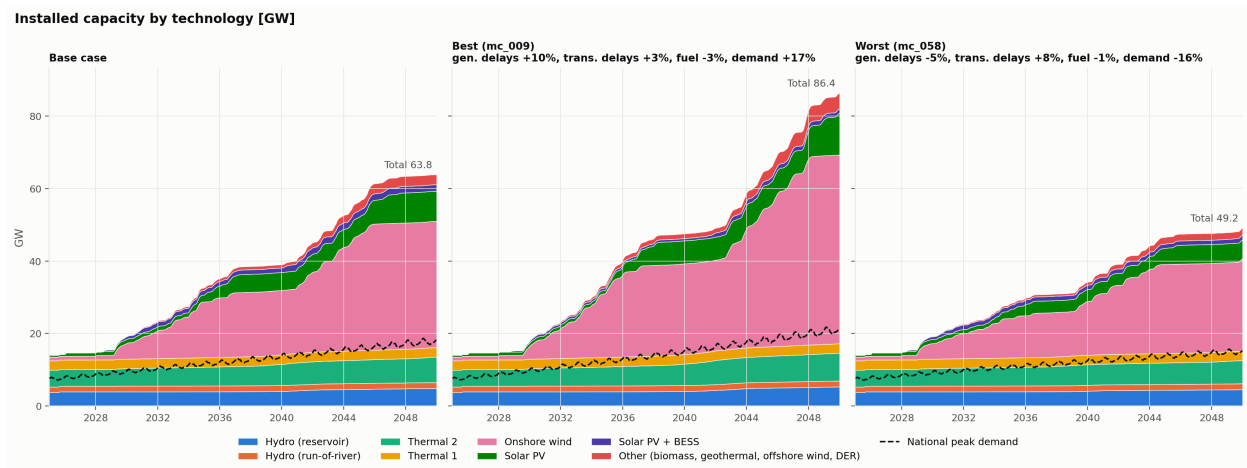


Figure 5: Installed capacity by technology for the central, best and worst case. The dashed line is national peak demand. Fourteen model technologies are aggregated into eight series; the mapping is given in Supplementary Material S5.3.

rather than fourth (Table 5): a longer pipeline defers reinforcements, which the series topology converts into persistent congestion rather than into a different final capacity.

That asymmetry explains why transmission lead time barely registers in the aggregate indicators while remaining operationally important. The model builds essentially the same network in the end; what the lead time changes is *when* it is available, and in a longitudinal system the cost of that timing shows up as congestion between zones rather than as a different system size.

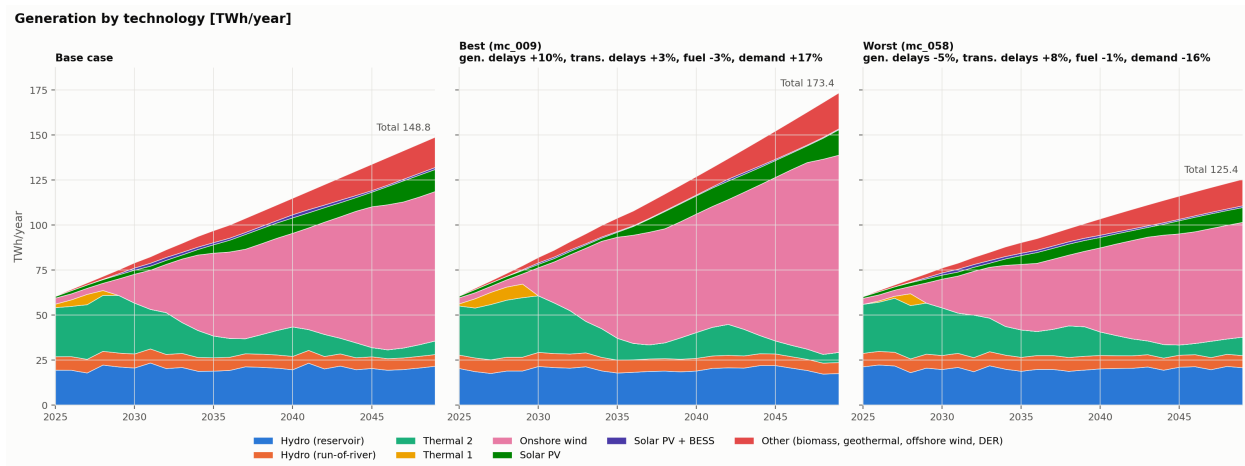


Figure 6: Annual generation by technology for the central, best and worst case.

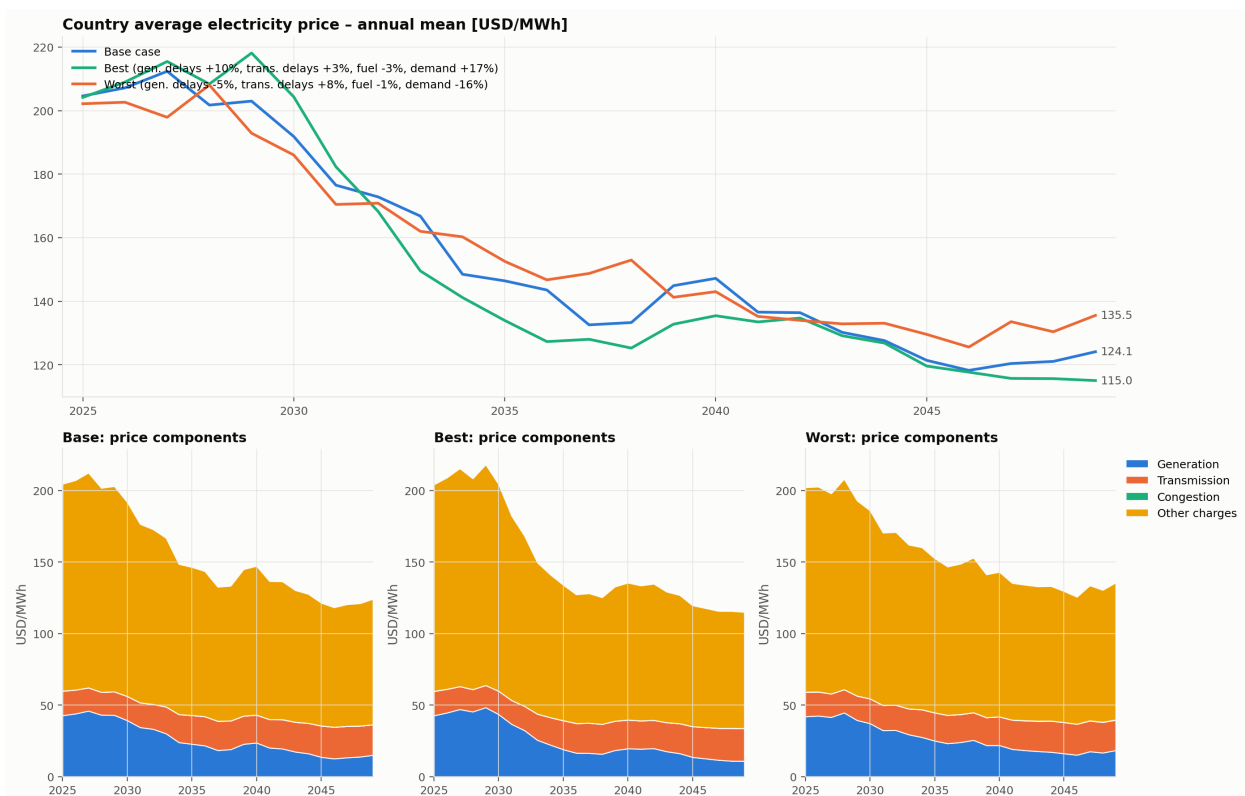


Figure 7: Country average electricity price for the central, best and worst case.

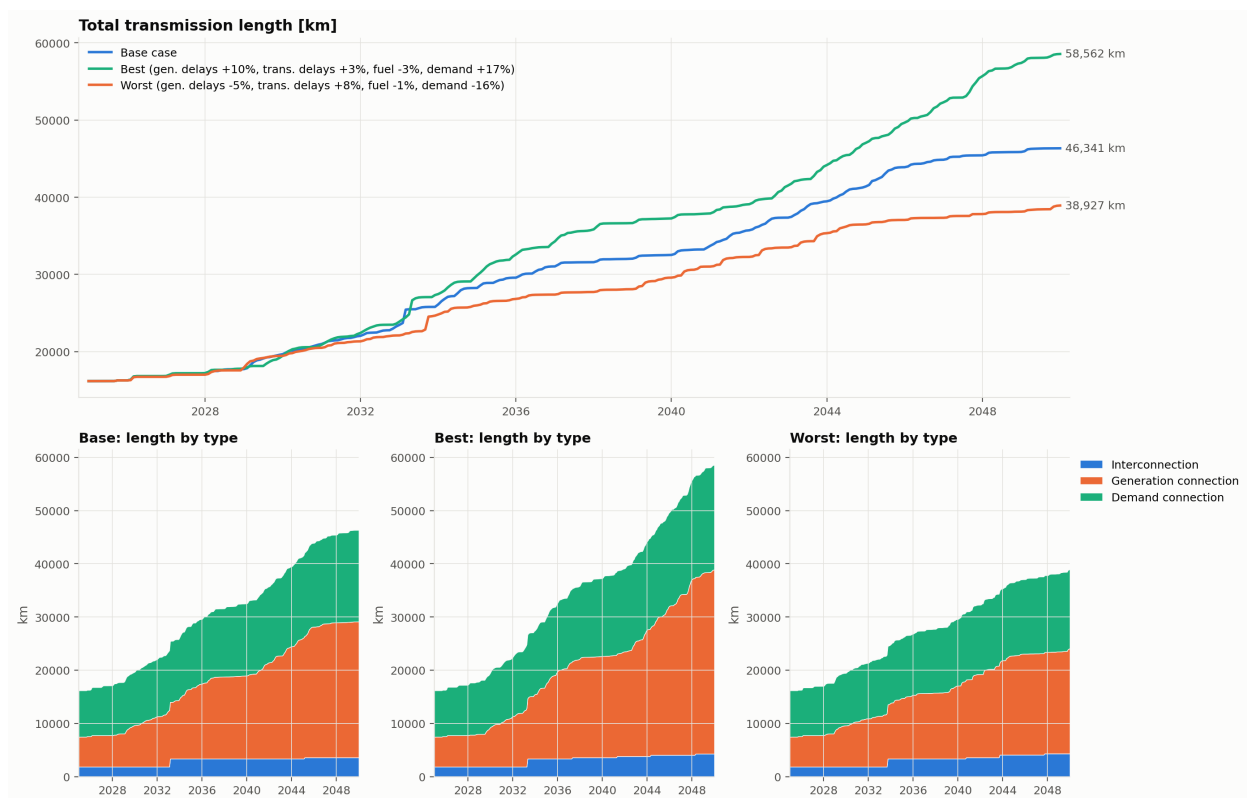


Figure 8: Transmission network by category — demand connection, generation connection and interconnection — for the central, best and worst case.

6. Discussion

6.1. Why transmission matters for Peru's energy transition

The results give the grid a specific and quantified role. Between 2025 and 2050 the central case requires the transmission network to grow from 16,180 km to 46,341 km — a factor of 2.9 — and the plausible range extends to 57,069 km at the ninetieth percentile. That requirement is not driven by a decarbonisation target; the model has none. It is driven by the locational mismatch described in Section 3.4: the capacity that the profitability rule selects is wind and solar, and those resources sit away from the centre zone that holds 60 % of demand.

This is the point at which Peru's longitudinal topology stops being a descriptive detail. In a meshed system, a corridor constraint has alternatives; in a system where the south reaches the rest of the country only through the centre, it does not. Herrera et al. [21] show for northeast Brazil that transmission constraints bound wind expansion directly, and Ritter et al. [37] show for Europe that delayed interconnection changes the final generation mix rather than merely postponing it. Our results are consistent with both, with the qualification that in Peru the binding quantity is the *rate* at which network can be delivered relative to the rate at which demand grows.

6.2. Demand growth as the governing uncertainty

That demand dominates is, in retrospect, structurally explicable. Demand enters the model at the root of every chain: it sets the adequacy trigger, it sets the residual demand that the dispatch clears and hence the price signal that drives the profitability trigger, and it sets the zonal peaks that drive demand-connection network directly. Fuel prices, by contrast, enter only at the dispatch, and in a system that ends up 85 % non-thermal in capacity terms, a cost shift on the shrinking thermal margin has little leverage. The rank correlation of -0.014 for fuel prices on the average price is not a modelling artefact; it is what one should expect once the marginal technology is no longer the one burning the fuel.

The policy reading is uncomfortable, because demand growth is the one driver on the list that planning does not control and forecasting handles poorly. It is also the driver for which the model's own range (± 20 % on the 2050 level) is arguably conservative for an emerging economy with incomplete electrification and potential industrial loads. Methods for long-horizon demand uncertainty [23, 1] are therefore not a peripheral refinement for Peruvian planning but the main lever on the quality of the answer.

6.3. The price result and what it does not say

The finding that faster demand growth lowers the long-run average price requires careful statement, because it is easy to over-read.

It is a statement about a *long-run average* under an investment rule that responds to profitability. Faster demand sustains the price signal long enough for low-cost renewable capacity to clear the threshold earlier and in greater volume; that capacity then depresses the marginal price. It is not a statement that demand growth is costless — the same runs require 26 % more transmission and 35 % more generation capacity, and the model prices neither the capital at risk nor the land and social cost of that build-out. Nor is it a statement about short-run prices: Figure 7 shows the high-demand case *above* the others through the late 2020s, before the capacity it induces arrives.

Three structural caveats bound it further. The model has no endogenous retirement, so thermal plant persists and the mix shifts by addition rather than by replacement. Storage and grid-enhancing technologies are disabled at the switch level in every run reported here, so the substitution between flexibility and network documented by Sousa et al. [46] and Sadeghian and Mohammadi-Ivatloo [39] is absent; enabling them would plausibly reduce the transmission requirement of the high-demand case and weaken exactly the

corollary we emphasise. And the dispatch uses one representative day per month, which under-represents sustained scarcity.

6.4. *Infrastructure delays and long-term reliability*

Lead times rank second and third in our design, well behind demand. We do not read this as evidence that delays are unimportant for Peru, and the distinction matters.

What the result says is that, over a $\pm 20\%$ band around current practice, lead times do not change the *size* of the system that gets built by 2050. What it does not say is that they do not change the system's condition along the way. Section 5.5 shows the mechanism: longer transmission pipelines defer reinforcements, and in a series topology the deferral surfaces as congestion between zones rather than as a different terminal capacity. An indicator measured at January 2050 is poorly suited to capture that, and this is a limitation of our reporting choice as much as a property of the system. A study aimed specifically at reliability would need congestion-hours, curtailment and unserved energy as indicators rather than terminal stocks.

There is also a range effect. We varied lead times by $\pm 20\%$ around the Peruvian base case, which is a plausible band for incremental administrative improvement. It is not the band relevant to institutional failure: the literature on delay causes — right-of-way disputes, licensing, local opposition [48, 6] — describes multi-year slippage on individual projects, well outside $\pm 20\%$ of a planned stage. Our ranking is conditional on the ranges in Table 4, as any sensitivity ranking must be.

6.5. *Thermal dependence and fuel-price exposure*

Peru's gas dependence is the feature most often identified as its principal vulnerability [38, 13, 9], and our results qualify that view in a specific way. Fuel prices are the weakest of the four drivers on every indicator, with a swing of $0.005 \text{ USD MW}^{-1} \text{ h}$ on the 2050 average price — effectively zero.

The qualification is temporal, not substantive. The model's own trajectory removes the exposure: by 2050 thermal plant supplies 7.5 TW h yr^{-1} of $148.8 \text{ TW h yr}^{-1}$ in the central case. Fuel-price risk is therefore a near-term exposure that the expansion path itself retires, provided the expansion path is delivered. If it is not — if the network cannot be built at the rate the investment rule assumes — the system remains thermal for longer and the exposure persists. Read that way, fuel-price risk in Peru is better understood as a *consequence* of transmission delivery than as an independent risk to be hedged separately, which is a different policy prescription from the one usually drawn.

6.6. *Renewable integration and transmission coordination*

The central case reaches 84.9% non-thermal capacity by 2050 with no renewable target, driven by profitability alone. That is a stronger statement about the competitiveness of Peruvian wind and solar than a target-constrained model could make, and it aligns with the optimisation-based findings of Fiestas-Chevez et al. [14] and De la Cruz Torres et al. [10] from a different methodological direction — which is worth noting, since agreement between an optimising and a behavioural model on the direction of the transition is more informative than either alone.

Where the two diverge is on what it costs to get there. An optimisation model coordinates generation siting and network expansion by construction; our model does not, because the Peruvian institutional arrangement does not. Transmission for generation is triggered by *confirmed* projects rather than anticipated ones (Section 4.2.2), which is reactive by design, and transmission lead times exceed generation lead times. Herding et al. [20] quantify the prize for closing that gap — around 10% lower total investment and up to 30% less network expansion under coordination — and Sauma and Oren [40] formalise the proactive-versus-reactive distinction that underlies it. Our results indicate that Peru is firmly on the reactive side, and that the cost of remaining there rises with demand growth.

Complementary options would attack the same constraint from the demand side. Distributed generation reduces the inter-zonal transfers a longitudinal grid must support [15, 54, 43], and grid-enhancing technologies raise the capability of existing corridors without new right-of-way [39]. Both are represented in the model but disabled in this experiment; quantifying them for Peru is the obvious next study, and our results indicate where the gain would be largest — the Z2–Z3 corridor under high demand.

6.7. Policy implications

Four implications follow, stated in the order our evidence supports them.

Transmission delivery capacity is the planning variable to protect. Demand growth governs the outcome and is outside the planner’s control; what the planner controls is whether the network can keep pace. Since the least-cost outcome is also the most network-intensive, the risk is asymmetric: under-building the grid forecloses the cheap trajectory.

Anticipatory network planning has a measurable value here. The reactive trigger plus the lead-time asymmetry is a structural bottleneck, not a parameter choice. Moving the trigger from confirmed to anticipated projects is a regulatory decision with a quantifiable payoff [40, 20].

Demand forecasting deserves the modelling effort that fuel-price hedging currently receives. Given a 1:900 ratio in rank correlation between demand and fuel prices for transmission requirements, the allocation of analytical attention in Peruvian planning is not proportionate to the sensitivities.

Restoring a predictable procurement signal matters more than the price it clears. The profitability trigger in the model responds to a sustained price signal. The suspension of the auction mechanism [8, 50] removes exactly that, and Del Río et al. [11] show that in Latin America the cost of capital rather than resource quality determines what is financed — a channel our model represents only through the discount rate.

6.8. Comparison with other Latin American systems

Peru’s three-zone series topology is unusual in the region and it drives the result. Brazil’s meshed system gives alternatives when a corridor binds, and the constraint appears as regional heterogeneity in deployment efficiency [44] or as curtailment [21], rather than as a single critical path. Chile’s system is also longitudinal and its curtailment problem is well documented; the comparison with Peru is therefore the more instructive one, and Reyes [35] note that Chile has moved considerably further from a comparable liberalised base.

The common regional thread is institutional. Guliev and Solovova [19] and Semperteguí [42] identify regulatory sovereignty rather than engineering as the barrier to Andean interconnection, and Boateng et al. [6] show institutional quality conditioning the response to otherwise identical incentives. Our finding that lead times matter less than demand should not be generalised to systems with weaker institutions or wider delay distributions than the $\pm 20\%$ band we sampled.

6.9. Methodological implications

Two points generalise beyond Peru.

First, pairing a system dynamics model with a designed experiment changes what the model can claim. A small set of named scenarios shows what a model does; it cannot separate a structural result from an artefact of one input assumption. Here the designed experiment turned a plausible narrative — that a gas-dependent system faces fuel-price risk — into a measured ranking that contradicts it. The cost is modest: 129 runs, mechanisable, with the design fixed before execution.

Second, reporting both one-at-a-time and all-factors-varying evidence is worth the redundancy. The two answer different questions and have complementary weaknesses, and their agreement (Tables 5 and 6) is

what makes the ranking defensible. Where they disagreed we would have to report it, and that disagreement would itself be a finding about interaction.

We would add a third, more practical observation. The infrastructure of this study — resumable execution, a design fixed and archived before any simulation ran, and explicit checks that each run actually produced data — was not incidental. Several of those checks caught silent failures during execution that would otherwise have entered the results as missing data rather than as errors. For simulation experiments of this size, that machinery deserves to be part of the method rather than left to each author’s practice.

7. Conclusions and policy implications

We set out to establish which uncertainty governs the trajectory of the Peruvian power system, rather than to forecast that trajectory. A system dynamics model with explicit generation and transmission pipelines, coupled to a congestion-aware dispatch and subjected to a 129-run designed experiment over 2025–2050, gives an unambiguous answer.

Demand growth dominates, by an order of magnitude.. Over the sampled ranges, demand growth reaches Spearman rank correlations of 0.985 with required transmission, 0.973 with installed capacity and -0.752 with the average price, while no other factor exceeds 0.27 in absolute value. A $\pm 20\%$ band on the 2050 demand level moves required transmission by -18.9% to $+23.8\%$ and installed capacity by -24.0% to $+29.3\%$. The one-at-a-time and the all-factors-varying evidence agree on this ranking.

Fuel-price risk is weaker than the literature on Peru implies, and for a structural reason.. Fuel prices are the weakest driver on three of four indicators, moving the 2050 average price by $0.005 \text{ USD MW}^{-1} \text{ h}$. The model’s own expansion path removes the exposure: thermal generation falls to 7.5 TW h yr^{-1} of $148.8 \text{ TW h yr}^{-1}$ by 2050. Fuel-price risk in Peru is therefore better understood as a near-term exposure contingent on transmission delivery than as an independent long-run risk.

The least-cost trajectory is the most network-intensive.. Faster demand growth lowers the long-run average price, because it sustains the price signal long enough to pull forward low-cost wind and solar. The best case on delivered price requires 58,562 km of transmission, 26.4 % more than the central case and 50 % more than the worst, with 10.3 % more inter-zonal transfer capability. For a system whose grid is already the binding constraint, the cheap path is the one that demands most of the planner.

Lead times change when, not how much.. Transmission lead times rank last or second-to-last on terminal indicators, but the decomposition by transmission category shows why: the model builds much the same network either way, and the deferral surfaces as inter-zonal congestion rather than as a different system size. Terminal stocks are the wrong indicator for that effect, which is a limitation of our reporting and a clear direction for the next study.

The expansion is renewable without being told to be.. The central case reaches 84.9 % non-thermal capacity by 2050 under a profitability rule alone, with no target, mandate or carbon price. That a behavioural model and the optimisation-based Peruvian literature [14, 10] agree on the direction of the transition, having disagreed on almost everything methodological, is itself worth recording.

Policy implications

The practical reading is that Peruvian planning attention is not proportionate to the sensitivities. Effort devoted to fuel-price exposure buys little; effort devoted to (i) long-horizon demand forecasting, (ii) the institutional capacity to deliver transmission at the rate demand requires, and (iii) moving the network trigger from confirmed to anticipated generation projects [40, 20], buys a great deal. Restoring a predictable procurement signal matters for the same reason: the investment response in the model is to a sustained price signal, and the suspension of the auction mechanism removes it.

Limitations

The ranking is conditional on the ranges in Table 4; a wider delay distribution, such as the multi-year slippage documented for individual projects elsewhere, could change it. Storage and grid-enhancing technologies are disabled in every run reported here, so the substitution between flexibility and network is absent and the transmission requirement of the high-demand case is, if anything, overstated. Retirements are exogenous, the dispatch uses one representative day per month, and the distributions are uniform ranges of admissibility rather than estimated densities. Each of these is documented with its consequence in the supplementary material.

Further work

Three extensions follow directly. First, enabling storage and grid-enhancing technologies and re-running the same design would quantify how much network the flexibility options actually displace — our results identify the Z2–Z3 corridor under high demand as where the gain would be largest. Second, replacing terminal stocks with congestion-hours, curtailment and unserved energy as reported indicators would capture the timing effect of lead times that our indicators miss. Third, the same design applied to Brazil and Chile with the same model structure would separate what is Peruvian from what is a property of longitudinal systems generally; the model and driver used here are already structured for that comparison.

Data and code availability

The Vensim model, the complete input workbook, the experimental design, the simulation driver, the post-processing code and the commented notebook that regenerates every figure in this paper are openly available at <https://github.com/sebastianzapatar/peru-expansion-uncertainty>. **[Replace with the DOI of the archived release before submission.]** Nothing required to reproduce the results is available only on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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